# Lateral and Vertical Transistors Using the AlGaN/GaN Heterostructure

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*(Invited Review Paper)*

***Abstract*— Power conversion losses are endemic in all areas of electricity consumption, including motion control, lighting, air conditioning, and information technology. Si, the workhorse of the industry, has reached its material limits. Increasingly, the lateral AlGaN/GaN HEMT based on gallium nitride (GaN-on-Si) is becoming the device of choice for medium power electronics as it enables high-power conversion efficiency and reduced form factor at attractive pricing for wide market penetration. The reduced form factor enabled by high-efficiency operation at high frequency further enables significant system price reduction because of savings in bulky extensive passive elements and heat sink costs. The high-power market, however, still remains unaddressed by lateral GaN devices. The current and voltage demand for high power conversion application makes the chip area in a lateral topology so large that it becomes more difficult to manufacture. Vertical GaN devices would play a big role alongside of silicon carbide (SiC) to address the high power conversion needs. In this paper, the development, performance, and status of lateral and vertical GaN devices are discussed.**

***Index Terms*— Gallium nitride (GaN), lateral transistors, vertical transistors.**

## I. Introduction

POWER conversion is ubiquitous in our everyday lives. Power conversion could mean stepping up or stepping down from one voltage level to another (boost or buck) or a conversion from dc to an ac voltage (inverter) or from one phase to three phases (phase converter), or just isolating from the supply line (power factor correction). The whole range of power for targeted application is shown in Fig. 1 [1]. With the increased dependence of us on electricity from data servers, mobile computing, and communications to hybrid cars, the role of power conversion is increasing at a rapid pace. At the heart of all power conversion is a switch.

A plot of the breakdown voltage ($V_{\rm bd}$) and on-resistance ($R_{\rm ON}$), the two important parameters for evaluating any material system as a switch, is shown in Fig. 2, where the field is assumed to be constant and ideal materials parameters are used. The two main losses in a switch are conduction and switching losses. Conduction loss ($I^2R_{\rm ON}$) decreases with lower $R_{\rm ON}$. Switching loss [$(I.V_{\rm op}\ (t_r+\ t_f)f$], where $I$ and $V_{\rm op}$ are the operating current and voltage, respectively, $t_r$ and $t_f$ are rise and fall times] increases with the switching frequency, $f$. The device loss (conduction plus switching) is minimized by optimizing the device area [2] at each frequency and the minimum device loss with frequency is thus obtained, as shown in Fig. 3 for Si, silicon carbide (SiC) and gallium nitride (GaN).

The losses with state-of-the-art Si-based solutions unfortunately are nearly 10% of the electrical energy generated in

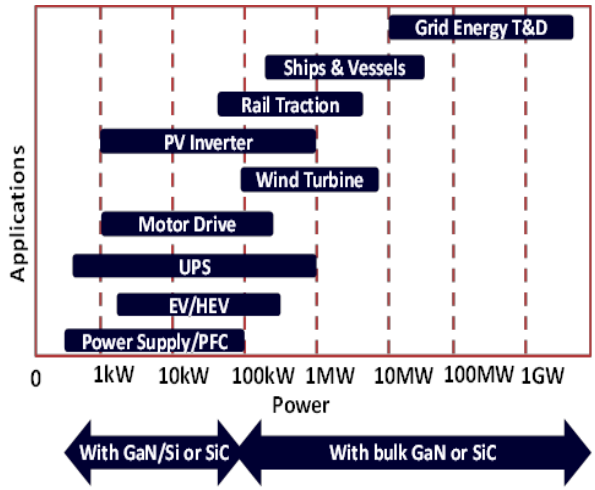



Fig. 1. Entire range of power applications that can be addressed with GaN (based on Yole Development report [1]).

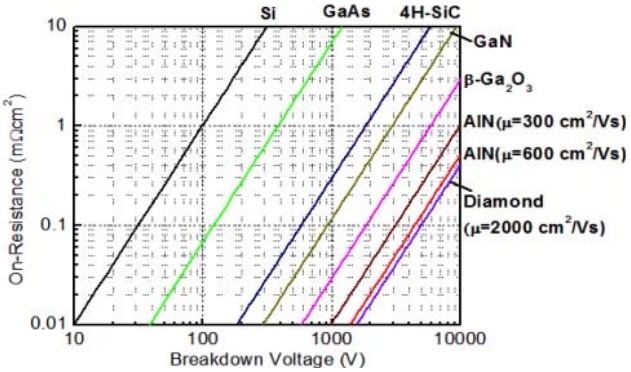



Fig. 2. On-resistance versus breakdown voltage for WBG materials; valid in 1-D (courtesy of Palacios Group, MIT).

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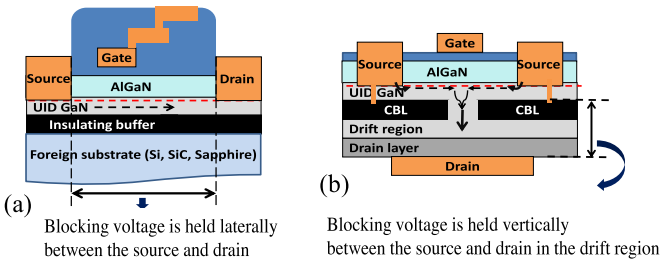



Fig. 3. (a) Lateral AlGaN/GaN power HEMT. (b) Vertical transistor using AlGaN/GaN layer structure on bulk GaN drift layer and substrate.

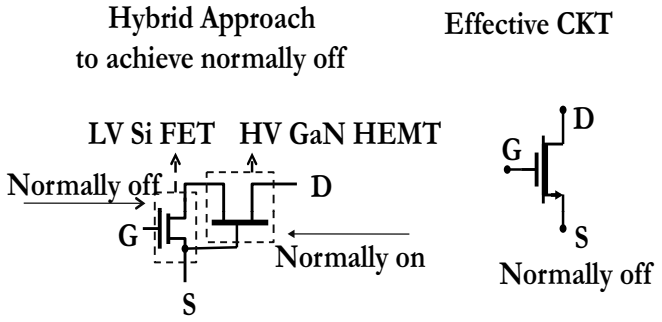



Fig. 4. Cascode normally off GaN HEMT allows for standard Silicon-based drive circuits while delivering the high voltage performance of GaN devices.

the United States and more than all the electrical energy generated by renewable energy sources combined. Wide bandgap (WBG) materials, promised, and have now delivered enhanced performance beyond the Si roadmap. The higher critical electric field ($E_c$) due to the large bandgap of these materials makes them ideal for high power electronics applications. In addition, lower device loss at higher frequencies makes WBG materials more attractive since they enable circuits operations at higher frequencies, thereby reducing the size of the passive components and making the system more compact and cost effective.

## II. Lateral and Vertical III-Nitride Transistors

Power devices in GaN can be either lateral devices on SiC and Si substrates or vertical devices preferably on GaN bulk substrates. Lateral devices are more mature and have demonstrated great promise [3]–[6]. Vertical GaN power devices [7] are attractive for high power applications when fabricated on bulk GaN substrates for the power range shown in Fig. 1. In this paper, we will describe the performance and promise of the AlGaN/GaN HEMT as the typical lateral device and the current aperture vertical electron transistor (CAVET) as the typical vertical device.

The AlGaN/GaN HEMT consists of an AlGaN layer grown on a GaN channel to take advantage of the high mobility (~2000 cm$^2$/V.s) 2-D electron gas (2-DEG) formed at the AlGaN/GaN interface as the current carrying layer. The source, drain, and gate are fabricated on the same plane on top [Fig. 3(a)]. The channel electrons are modulated typically by a MIS gate to limit gate leakage, and the distance between the gate and drain sustains the blocking voltage in the off-state of the device. Field plate structures [shown connected to the gate in Fig. 4(a)] with proper passivation (typically SiN) are designed to reduce the peak electric field between the gate and the drain region by imaging the depletion charge on a larger area (the field plate) than the active gate. The reduced electric field also prevents electron injection into traps, thereby preventing current collapse in these devices. However, for higher power (>50 kW) application where higher breakdown voltages (>1.2 kV) are required, the lateral topology becomes increasingly unattractive both in cost and manufacturability, due to very large chip areas required by the breakdown voltages at the desired operating current level (typically over 50 A). The power level at which vertical devices become preferred over lateral devices is currently under investigation. A typical vertical device presented as a CAVET [Fig. 3(b)] has source and gate on the top and the drain at the bottom. The current controlled by the gate, flows through the bulk of the material into the drain through an aperture between current blocking layers (CBLs) typically achieved by either p-type doping of GaN layer or by isolation implantation. The horizontal high mobility electron channel achieved by the AlGaN/GaN layer is used in conjunction with a thick GaN drift region to achieve low $R_{ON}$ and high breakdown voltage. Since vertical devices sustain the blocking voltage in the vertical direction into the bulk material of the device, it makes the chip area smaller for a specific current of operation compared with the lateral design. The high electric field region being buried in the bulk also alleviates current collapse due to surface traps, eliminating or reducing extensive passivation and field plating as required by the lateral designs.

In the next sections, we will discuss at first the more mature lateral devices followed by the vertical transistors structures.

## III. Lateral GAN HEMT Normally off Solution

Power electronic applications demand normally off devices. There are several methods of achieving normally off operation.

1) Junction gated devices to pinchoff the channel under the high work function gate whereas maintaining the channel in the access regions that do not have the p region have been implemented by EPC and Panasonic, the latter as the gate injection transistor [8].
2) A multichannel transistor with two AlGaN/GaN interfaces has been developed by Toshiba, Fujitsu, NTT, and Transphorm [9]. Here, a conductive access region is defined by a 2-DEG formed at the upper AlGaN/GaN interface (as in a normal HEMT). A second channel below is accessed via gate recess etching and MIS gate formation within the recess. The second channel is formed between a thin AlGaN layer and GaN such that there is no charge in the channel at zero bias and charge is induced under forward bias (normally OFF).
3) Normally off operation by introducing $F$ under the gate structure to shift the threshold voltage positive under the gate while maintaining a conductive channel in the access regions has been developed by HKUST [10].
4) Finally, the channel etch through structure MOSFET where the 2-DEG is etched through and an MOS gate deposited has been developed by RPI [11].

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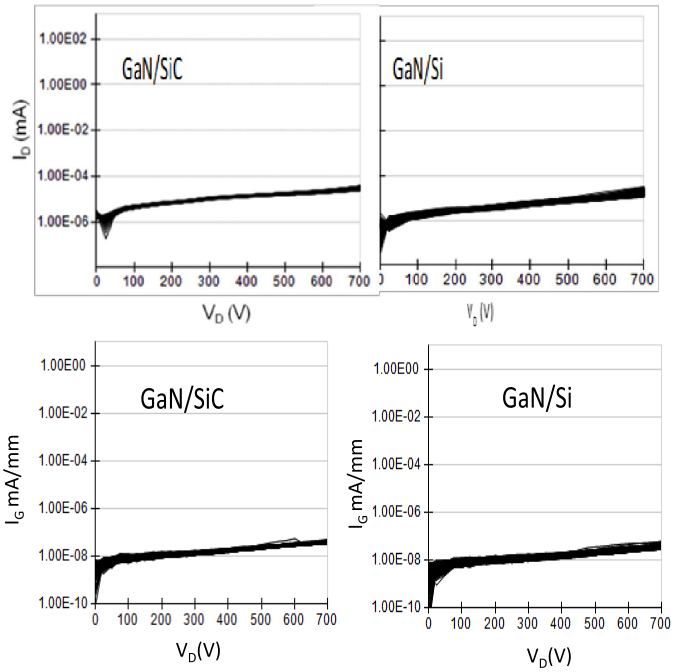



Fig. 5. GaN/Si and GaN/SiC HEMT leakage (drain and gate) comparison data shows minimal impact of disclocations/GaN-Si material on achieving comparable and lower leakage than GaN on SiC devices.

The advantage of all these structures is that it is a single chip solution. The disadvantage of the MIS gated devices is that the gate insulator on GaN and AlGaN is not well developed and the reproducibility of the interface properties and stability under operation needs to be significantly improved. The junction-gated devices have low threshold voltage and limited gate-bias drive range because of the turn-on of the p-n junction gate.

The simplest embodiment so far of normally off GaN transistors incorporates a cascode approach at the packaged level. As shown in Fig. 4, a normally off low-voltage Si FET is connected to a normally on high-voltage GaN HEMT in series while the gate of the GaN HEMT is connected to the source of the Si FET. This hybrid configuration produced an effective e-mode power device, safe in case of faulty gate control. Moreover, it provides the compatibility with existing Si drivers, as well as the freedom to optimize the GaN HEMTs without the complication of special gate drive circuits.

## IV. DC HEMT and Cascode HEMT Performance

GaN epi-layers were grown on 4"–6" SiC and Si substrates. These epi films were developed to have a low defect density ($5 \times 10^8$–$1 \times 10^9$) cm$^{-2}$ on SiC and ($2 \times 10^9$–$1 \times 10^{10}$) cm$^{-2}$ on Si with a high 2-DEG mobility and high charge density such that both a high breakdown field and a low channel resistance are achieved. The device structure consists of a resistive (Al, Ga)N-based buffer layer capable of sustaining the rated voltage and the design of which is such that the final wafer has $< 20$-$\mu$m bow. Depletion mode GaN-based HEMTs with insulated gates were then fabricated. The finished wafers were characterized with auto probers of high current and high voltage capabilities. The typical on-wafer

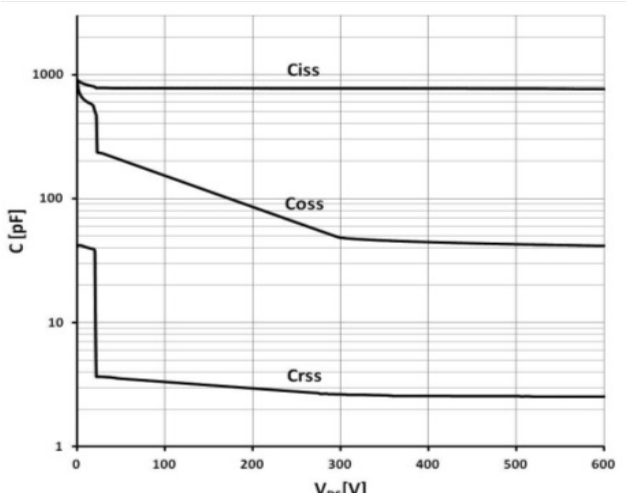



Fig. 6. Capacitance versus voltage plot for transphorm's 600 V HEMT product TPH3006.

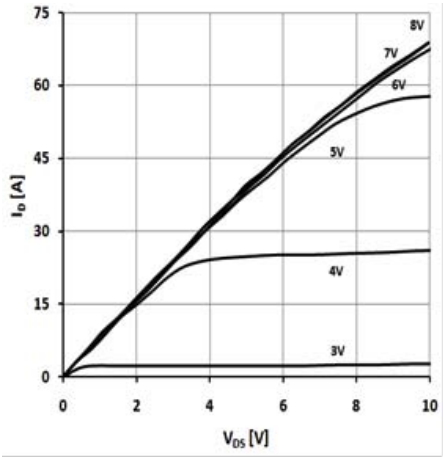



Fig. 7. $I$–$V$ curves for transphorm's 600 V HEMT product TPH3006.

off-state drain leakage $I_D$ and gate leakage $I_G$ of 120-mm-wide GaN-on-Si and GaN-on-SiC HEMTs are shown in Fig. 5. At $V_D = 600$ V, the room temperature drain leakage is as low as 10–50 nA and gate leakage is 3–15 nA. At 150 °C, the typical drain leakage increased to 1–10 $\mu$A and gate leakage is ~0.1–1 $\mu$A at 600 V.

Contrary to popular myth, dislocations do not impact the leakage current levels in otherwise well designed and optimally manufactured lateral GaN devices, as evidenced by the similar leakage performance of our GaN on Silicon HEMTs compared with our GaN on SiC HEMTs (Fig. 5). The typical dc on-resistance of 120-mm HEMT is ~0.1–0.11 Ω and the capacitance ($C_{oss}$) is 40 pf (Fig. 6).

The plastic-packaged GaN HEMT has a gate threshold of +2.1 V typical at 1-mA drain current. The dynamic ON-resistance is 0.15-Ω typical and 0.18-Ω maximum. The pulsed drain current is 70 A at a $V_G = 8$ V and $V_D = 10$ V. The continuous-wave drain current is 14 A at a case temperature of 25 °C (Fig. 7).

The $R_{ON}$ ranges from 250 to 150 mΩ based on device size. Comparing with similarly rated state-of-the-art Si super-junction MOSFETs on the market, the GaN devices offer

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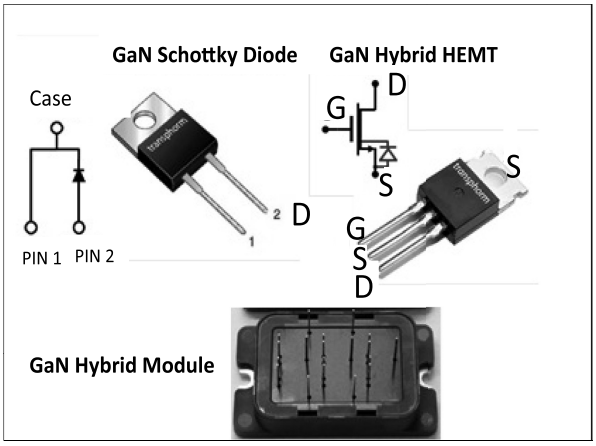



Fig. 8. GaN diode, transistor, and module products.

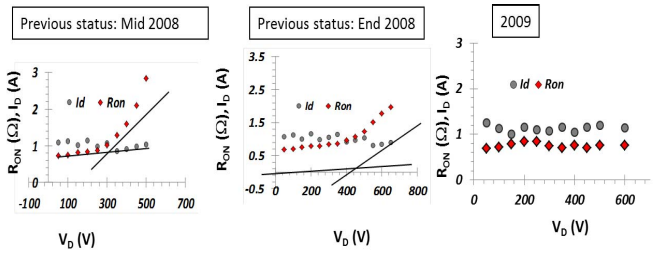



Fig. 9. History of reduction of dynamic on-resistance for GaN HEMTs at transphorm. It is important to achieve no increase in on-resistance at ac conditions to realize high performance in applications.

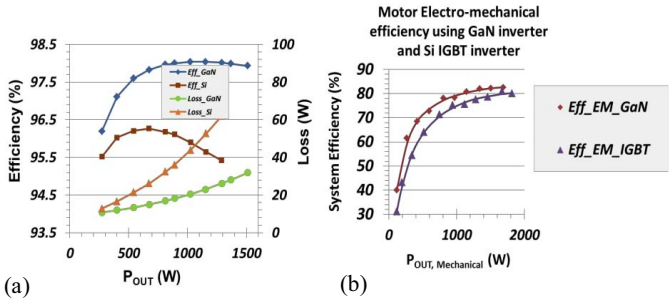



Fig. 10. (a) and (b) GaN solutions enable high frequency (high density) operation of Boost/PFC with 70% loss reduction over silicon super-junction. 6:1 GaN modules with integrated filter show enhanced electromechanical efficiency by making motors more efficient using pure sinusoidal drive signals.

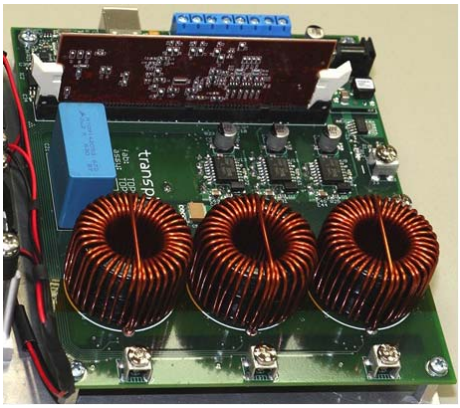


Fig. 11. Compact filter is fitted on the same board as the inverter to filter out the higher harmonics of the PWM signal.

significant reduction in gate charge, on-resistance, and output capacitance. The lateral GaN device in a TO-220 package also features the Quiet-Tab package scheme, with gate-source-drain pin-out arrangement and the package base as a low-inductance source terminal. This configuration allows 200% increase in switching speed compared with traditional TO-220 packages, which is necessary to take full advantage of the low switching losses of GaN. The GaN diode has a Schottky barrier with a forward voltage of 1.3 V at 2-A operating current depending on size, with no minority or reverse recovery charges. The packaged devices are shown in Fig. 8. Only GaN transistors will be addressed in this paper.

As any well-designed device with safety margin, our 600 V rated devices have breakdown voltages in excess of 900 V. We have also tested the robustness of these devices against spikes, by intentionally subjecting them to 900 V spikes (1-million spikes) in a true switching environment. No performance degradation is observed. While the present breakdown voltage is much lower than the inherent avalanche capability of GaN material, the design margin available in our devices guarantees against failure in appropriately designed end user circuits and applications.

Good dispersion control is the key to achieve the low on-resistance in switching operation. Small devices were used for on-wafer evaluation of dynamic performance. The increase of dynamic $R_{ON}$ at 500 V compared with its static value is usually 10%–20%. The dynamic performance is again verified in final packaged devices, guaranteeing that users of this device will indeed match up in-circuit performance with datasheet values of resistance. The dynamic $R_{ON}$ of packaged device measured with low duty cycle at various prebias voltages is shown in Fig. 9. It increased only 0.01 Ω from 0 to 200 V and then saturated up to 600 V. The device is measured at 50-Hz repetition frequency. The device is on for 2.5 $\mu$S/ cycle. The dynamic $R_{ON}$ was measured at 2 $\mu$S into the on-state.

Fig. 10(a) shows a 750 W, 230–400 V boost converter with the total-GaN solution (using both GaN HEMT and GaN diode), which showed cleaner and faster waveforms as well as 33% and 70% reductions of device losses at 100 and 500 kHz, respectively, compared with a converter with the state-of-the-art Si transistor and Si diode in the market. The hybrid GaN HEMT is also capable of three-quadrant operation and has 25-time less reverse recovery charge than the best Si super-junction MOSFETs. This makes diode-free hard-switched bridge possible. 600 V six-in-one power modules with on-resistance of 150-mΩ/switch were developed for operation at 100–300 kHz, about 10 times higher than traditional power modules. The high pulsewidth modulation (PWM) frequency allows integration of compact output filters resulting in a three-phase pure-sine-wave inverter for PV or motor drive application (Fig. 11). An actual motor operation test at 100-kHz PWM has revealed significant electromechanical efficiency boost by the GaN inverter by 8%, 4%, and 2% at low, mid, and high load, respectively, compared with a state-of-the-art IGBT inverter at 16 kHz [Fig. 10(b)] [12].

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## V. 600 V GaN-on-Silicon Diode and Transistor Qualification

Transphorm has completed the industry's first qualification of 600 V GaN on Silicon devices in 2013. The qualification testing followed JEDEC standards and comprised of accelerated stress and reliability testing on three lots of 77 parts each, per JESD-47 standard specification. The requirement to pass is zero fails across the lots. Specific testing included.

1) Electrical tests.

a) HTRB: $T$j = 150 °C, 480 V, min: 1000 h;

b) HAST: 130 °C/85% relative humidity, 33.3 psi, continuous bias at 100 V, min: 96 h;

c) Temperature cycle: −55 °C/150 °C, cycles/h, min: 1000 cycles;

d) Power cycle: 25 °C/150 °C, $\Delta T$ = 125 °C, min: 5000 cycles;

e) HTSL: 150 °C, min: 1000 h.

2) Electrostatic discharge tests (charged device model, machine model and human body model).

3) Mechanical testing including destructive physical analysis, wire pull, ball shear and die shear, vibration, shock, and visual verification.

All device parameters that are on the device data sheet are met even after the stress tests are performed.

## VI. Vertical Transistors in III-Nitride

Vertical GaN device research is in its infancy compared with the work on lateral devices. The prominent device to date has been the CAVET.

The CAVET was reported in 2000 by Yaacov *et al.* [13] and originally designed for RF power application for dispersion-free performance, exploiting the advantage of the high field region being buried under the gate within the bulk of the material. Later in 2009, the device was redesigned and reported as a high voltage switch for power electronics application and demonstrated a very promising depletion and enhancement mode performance [14]. The breakdown voltage was improved to 200 V as reported in 2012, with a 3-$\mu$m drift region and $R_{ON}$ < 2.2 mΩ cm$^2$ [15]).

Kanechika *et al.* [16] reported a vertical insulated gate HFET, which is similar to the CAVET, on bulk GaN substrates in 2007 with $R_{ON}$ = 2.6 mΩ cm$^2$.

Also in 2007, Otake *et al.* [17] reported vertical MOSFETs on bulk GaN with a threshold voltage of 3.7 V and $R_{ON}$ of 9.3 mΩ cm$^2$.

The following section will discuss CAVET device structure, process steps, and performance in detail.

## VII. Operation Principle, Fabrication, and Results from CAVET

### A. Operation Principle

The device (see Fig. 12) comprises of two source regions on either side of the gate on the top and drain at the bottom. The electron channel is comprised of a high mobility 2-DEG layer at an AlGaN/GaN interface. Electrons flowing laterally through the channel are modulated by planar gates and then

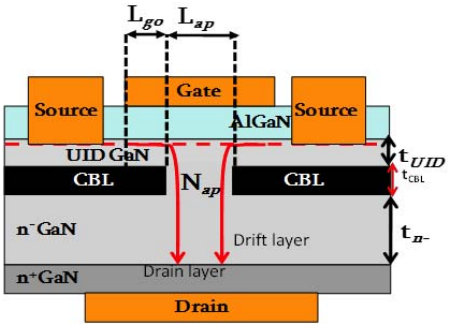



Fig. 12. Schematic layout of a CAVET with all the critical parts of the device labeled.

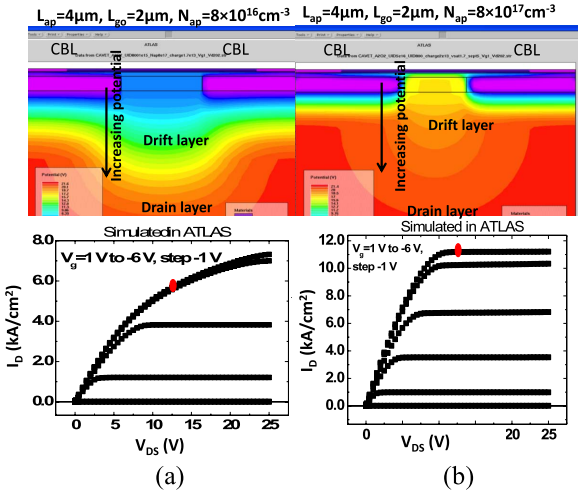



Fig. 13. (a) CAVET with resistive aperture design showing unsaturated device $I_d$-$V_{ds}$ characteristics. (b) Same CAVET with higher doped aperture does not choke the current, thereby leading to saturation. Simulation done in Silvaco Atlas.

directed vertically through a conductive aperture layer into an n-GaN drift region designed to hold the blocking voltage into the bulk of the device. The n$^+$ substrate forms the drain contact. CBLs are placed appropriately to prevent current from flowing vertically into the drain from any other region in the device other than the aperture. The conductance of the aperture is an important design parameter and is given ($L_{ap}.q\,N_{ap}.W_g/t_{CBL}$) is higher than the channel resistance (where $L_{ap}$ = is the length of the aperture, $q$ is the electron charge, $N_{ap}$ the doping of the aperture region, $W_g$ is the gate width of the device, and $t_{CBL}$ is the thickness of the CBL region). A simulation, done in Silvaco ATLAS, shows the effect of the aperture conductance on the saturation of device current with drain voltage. With a resistive aperture, the drain current does not saturate at low drain bias, as shown in Fig. 13(a) since the resistive aperture region considerably holds the applied voltage. Increasing the aperture conductance will decrease the voltage drop in the aperture to a very low value rendering the CAVET to be two lateral devices in parallel, which is also undesirable. Optimizing the aperture conductance leads to the desirable outcome of both saturation of drain current and redistribution of the voltage in the bulk, as shown in Fig. 13(b). Unlike lateral HEMTs, the high electric field region under the OFF-state operation is buried under

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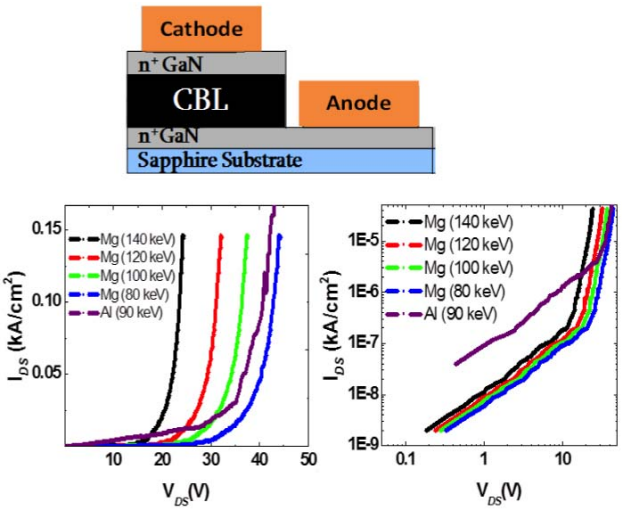



Fig. 14. (a) Two terminal structure to test the blocking capability of the CBL. ( b) $I$–$V$ characteristics obtained from simple two terminal measurements, as shown in (a) showing that the Mg implanted CBL bolocking is superior to Al implanted blocking layer. The optimum energy was found to be 80 keV. (c) Log $I$-log$V$ plotted for the same device.

the gate in a CAVET, and therefore the surface state-related dispersion is alleviated in these devices.

### B. Fabrication Challenges of a CAVET

The design space in a CAVET involves designing a good CBL and an aperture with optimum conductance. In the simulations done in Silvaco ATLAS, a p-type GaN layer was used as the CBL as it provides a very effective CBL with an electron barrier of over 3 eV. As-grown Mg-doped layers by metal–organic chemical vapor deposition (MOCVD) are typically passivated by $H$ and hence not active. Activating Mg in a buried GaN layer has been, however, found always challenging in [13] and the estimated energy barrier has never been achieved. Another way of achieving the CBL is by ion-implantation. Fabrication of a CAVET involves a two-step process where the channel and the cap above the aperture are regrown after the CBL is formed in the desired regions. In this paper, done at UCSB, Mg was chosen as the preferred implantation species with chosen dose of $10^{15}\text{cm}^{-2}$ and implantation energy of 80 keV, at RT.

This dose was chosen after a study of the conduction in two terminal structures where the Mg: implanted CBL was sandwiched between two $n^+$GaN layers, which formed the anode and cathode contacts [see Fig. 14(a)]. Under bias, the electrons from the cathode (grounded) flow into the anode held at a positive potential, through the CBL. This resembles the off-state scenario of a CAVET where the drain is subjected to a positive voltage (like the anode) and the source is grounded (similar to the cathode). The difference is that the two terminal structures do not have the thick drift region present in a CAVET to absorb most of the applied voltage and hence represents a more stringent test that would be observed under normal operating conditions in a CAVET. A comparison with Al as an isolation species [Fig. 14(b) and (c)] showed that Mg had a better insulating property and hence was preferred.

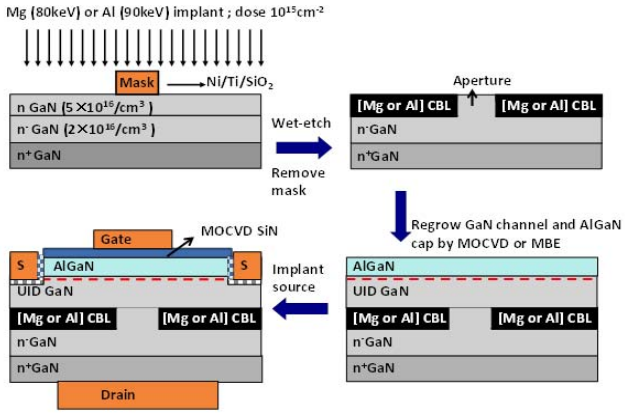



Fig. 15. Schematic representation of the process flow of a CAVET on bulk GaN substrate.

### C. DC Behavior of the CAVET

The process flow of a CAVET is shown in Fig. 15. The drift region of the CAVET was grown by Toyota Motor Corporation, Japan, which consisted of a 0.3-$\mu$m-thick GaN (Si: $5 \times 10^{16}$ cm$^{-3}$), 3-$\mu$m-thick n-GaN (Si: $2 \times 10^{16}$ cm$^{-3}$) on $n^+$GaN substrate. The CBL was implanted selectively around the aperture with $10^{15}\text{cm}^{-2}$ [Mg] at 80 keV. Regrowth of the GaN channel and the $Al_{0.3}Ga_{0.7}N$ cap was achieved using plasma MBE growth at 700 °C. This was to avoid redistribution of the implanted Mg during the high temperature MOCVD regrowth (1165 °C), which led to uncontrolled threshold voltage shift in devices. More details of the regrowth technique in both MOCVD and MBE systems have been discussed elsewhere [14], [15]. Implanted source contacts were used to contact the source ohmic region and the drain contact was formed at the back of the conductive GaN substrates. Nonalloyed contacts to the implanted source regions were necessary to prevent the commonly observed spiking of alloyed metals from creating a short through the CBL. Device to device and bond pad isolations were obtained by dry etching (reactive-ion etching) of mesa regions and covering the trenches with an MOCVD-deposited SiN layer. Ni/Au/Ni gate was laid on top of a 25-nm-thick MOCVD deposited SiN layer used as the gate dielectric to reduce the gate leakage in the device. The CBL was shorted to the source to assist easy charging and discharging of the CBL under operation.

The results obtained from CAVETs with Mg implanted CBL using MBE regrowth showed promising $R_{\text{ON}}$ (2.2 m$\Omega$ cm$^2$ and $V_{\text{BD}}$ (200 V). The $I_{d\text{ON}}/I_{d\text{OFF}}$ of this device was $\sim 5 \times 10^4$. This ratio is determined primarily by the gate-drain overlap ($L_{\text{go}}$), which serves as the effective gate length of the device. Dependence of $R_{\text{ON}}$ on the $L_{\text{ap}}$ for the same doping density and the same aperture to gate overlap optimized for minimal parasitic leakage is shown in Fig. 16 (see Fig. 12 for the device schematic). $R_{\text{ON}}$ decreases with an increase of $L_{\text{ap}}$ since the conductance in the aperture increases proportional to $L_{\text{ap}}$. A saturation in $R_{\text{ON}}$ is observed when the dominant part of the resistance is offered by the channel and drift region. Using an increased aperture width beyond the value that gives the saturated value of $R_{\text{ON}}$ is not desirable as it increases

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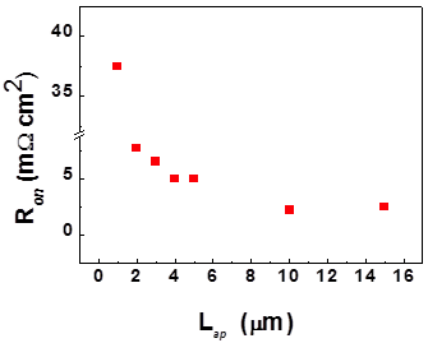



Fig. 16. Ron decreases with increasing $L_{ap}$ ($N_{ap}$ and $L_{go}$ are kept constant at $5 \times 10^{16}$ cm$^{-3}$ and 3 $\mu$m, respectively). This was obtained from the MBE regrown CAVET with Mg-implanted CBL.

the chip area. The MBE regrown CAVETs did not show any current collapse suggesting that a lower regrowth temperature is favored to keep the defects localized within the CBL. The breakdown in these devices was observed in the gate dielectric layer suggesting that the measured breakdown electric field of 0.7 MV/cm was limited by the dielectric material and not the bulk GaN.

## VIII. Conclusion

The first generation of lateral GaN devices has already been JEDEC qualified and has shown superior performance to state-of-the-art Si CoolMOS and IGBT solutions in power supply and inverter applications. The continuous enhancement of device design and new applications enabled by the unique properties of GaN, high dc, and switching performance coupled with the bidirectional nature of the switch will lead to ever better device metrics and ever expanding applications. The development of vertical transistors based on GaN will complete the device portfolio for the complete set of applications (from low to high power conversion) discussed in Section I. The speed of development of vertical devices will be accelerated with the development of GaN substrates where continuous breakthroughs are on-going. A disruption in power conversion using GaN-based solutions is upon us.

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